Real Numbers and Rational Exponents

Concept

Real numbers include rational and irrational numbers. Fractional powers represent roots.

Classification

Rational: integer fractions; irrational: non-repeating, non-terminating decimals.

Fractional power

$a^{m/n} = (\sqrt[n]{a})^m$ for $a > 0$; reduce m/n first.

Negative power

$a^{-m/n} = 1/a^{m/n}$, on its non-zero real domain.

Root sign

$\sqrt{(x^2)} = |x|$; odd roots can accept negative inputs.

Example 1

Evaluate $32^{3/5}$ and $81^{-1/2}$.

Solution

1. $32^{3/5} = (\sqrt[5]{32})^3 = 2^3 = 8$.

2. $81^{-1/2} = 1/\sqrt{81} = 1/9$.

Real Numbers and Rational Exponents

Example 2

Simplify $\sqrt{(49x^2)}$, then evaluate it at $x = -2$.

Solution

1. $\sqrt{(49x^2)} = 7|x|$, because the square root is non-negative.

2. At $x = -2$: $7|-2| = 14$.

Real Numbers and Rational Exponents

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $64^{2/3}$.

2. Evaluate $25^{-3/2}$.

3. Simplify $\sqrt{(49x^2)}$.

4. For $x > 0$, simplify $x^{5/6}/x^{1/3}$.

5. Is $\sqrt[3]{(-8)}$ real? Is $\sqrt{(-8)}$ real?

Real Numbers and Rational Exponents

Practice answers | check after trying

1. $(\sqrt[3]{64})^2 = 4^2 = 16$.

2. $1/(\sqrt{25})^3 = 1/125$.

3. $7|x|$.

4. $x^{5/6 - 2/6} = x^{1/2}$.

5. $\sqrt[3]{(-8)} = -2$ is real; $\sqrt{(-8)}$ has no real value.

Radical Operations and Equations

Concept

Simplify radicals by extracting perfect powers. Check equations after squaring.

Like radicals

Combine only matching radicals after simplification.

Domain

A real square root needs a non-negative radicand.

Rationalize

Multiply numerator and denominator by a radical or conjugate.

Equation steps

Restrict → isolate radical → square → solve → check.

Example 1

Simplify $\sqrt{50} + 2\sqrt{8} - \sqrt{18}$.

Solution

1. Extract squares: $5\sqrt{2} + 4\sqrt{2} - 3\sqrt{2}$.

2. Combine: $6\sqrt{2}$.

Radical Operations and Equations

Example 2

Solve $\sqrt{(x + 6)} = x$.

Solution

1. The right side must be non-negative: $x \geq 0$.
2. Square: $x + 6 = x^2 \rightarrow (x - 3)(x + 2) = 0$.
3. Candidates: 3 and -2. Check original: only $x = 3$ works.

Example 3

Rationalize $1/(\sqrt{3} + 1)$.

Solution

1. Multiply by $(\sqrt{3} - 1)/(\sqrt{3} - 1)$.
2. Result $= (\sqrt{3} - 1)/(3 - 1) = (\sqrt{3} - 1)/2$.

Radical Operations and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify $\sqrt{108}$.

2. Simplify $3\sqrt{12} - \sqrt{27}$.

3. Rationalize $5/\sqrt{2}$.

4. Solve $\sqrt{(2x + 3)} = 5$.

5. Solve $\sqrt{(x + 2)} = x$.

Radical Operations and Equations

Practice answers | check after trying

1. $6\sqrt{3}$.

2. $6\sqrt{3} - 3\sqrt{3} = 3\sqrt{3}$.

3. $5\sqrt{2}/2$.

4. $2x + 3 = 25$ gives $x = 11$; it satisfies the original.

5. $x \geq 0$. Squaring gives $x^2 - x - 2 = 0$, with candidates 2 and -1. Only 2 satisfies $\sqrt{4} = 2$; -1 is extraneous.

Advanced Polynomial Factoring

Concept

Use structure to factor: common factors, trinomials, grouping and square differences.

GCF first

Remove a common factor before choosing another pattern.

Split the middle

For $ax^2 + bx + c$, find product ac and sum b.

Patterns

$A^2 - B^2 = (A - B)(A + B)$; $A^2 \pm 2AB + B^2 = (A \pm B)^2$.

Grouping

Factor a shared binomial; substitute u for repeated expressions if useful.

Example 1

Factor $6x^2 + 11x + 3$.

Solution

1. $ac = 18$; choose 9 and 2 because $9 + 2 = 11$.
2. $6x^2 + 9x + 2x + 3 = 3x(2x + 3) + (2x + 3)$.
3. Result: $(3x + 1)(2x + 3)$.

Advanced Polynomial Factoring

Example 2

Factor $12x^3 - 75x$ completely.

Solution

1. GCF: $3x(4x^2 - 25)$.

2. Difference of squares: $3x(2x - 5)(2x + 5)$.

Advanced Polynomial Factoring

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Factor $3x^2 + 10x + 8$.

2. Factor $2x^2 - 7x + 3$.

3. Factor $18x^2 - 8y^2$.

4. Factor $x^2 - 6x + 9$.

5. Factor $5x(x - 2) + 3(x - 2)$.

Advanced Polynomial Factoring

Practice answers | check after trying

1. $(3x + 4)(x + 2)$.

2. $(2x - 1)(x - 3)$.

3. $2(9x^2 - 4y^2) = 2(3x - 2y)(3x + 2y)$.

4. $(x - 3)^2$.

5. $(x - 2)(5x + 3)$. Expand each product to verify all coefficients.

Rational Expressions and Equations

Concept

Rational expressions are polynomial fractions. Original denominators determine restrictions.

Cancel factors

Never cancel separate terms of a sum.

Add / subtract

Use a common denominator, then combine whole numerators.

Divide

Multiply by the reciprocal; the divisor must also be non-zero.

Solve

Record exclusions → multiply by LCD → solve → reject invalid roots.

Example 1

Simplify $(x^2 - 4)/(x^2 + 3x + 2)$.

Solution

1. Factor: $(x - 2)(x + 2)/[(x + 1)(x + 2)]$.
2. Restrictions: $x \neq -1, -2$. Cancel factor $x + 2$.
3. Result: $(x - 2)/(x + 1)$, keeping both restrictions.

Rational Expressions and Equations

Example 2

Solve $(x + 1)/(x - 1) = 2/(x - 1)$.

Solution

1. Restriction: $x \neq 1$.

2. Multiply by $x - 1$: $x + 1 = 2 \rightarrow x = 1$.

3. Reject excluded value 1. There is no solution.

Rational Expressions and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify $(x^2 + 5x + 6)/(x + 2)$.

2. Subtract $3/x - 1/(2x)$.

3. Solve $3/(x - 2) = 1$.

4. Solve $x/(x + 1) = 1$. State restrictions in each case.

Rational Expressions and Equations

Practice answers | check after trying

1. $x + 3$, $x \neq -2$.

2. $6/(2x) - 1/(2x) = 5/(2x)$, $x \neq 0$.

3. $x \neq 2$; $3 = x - 2$ gives $x = 5$, which checks.

4. $x \neq -1$; $x = x + 1$ is impossible, so there is no solution.

Quadratic Functions and Transformations

Concept

A quadratic graph is a parabola. Its equation form reveals different features.

Vertex form

$y = a(x - h)^2 + k$; vertex (h,k), symmetry $x = h$.

Opening / range

$a > 0$: $y \geq k$; $a < 0$: $y \leq k$, on the full real domain.

Other forms

$ax^2 + bx + c$ shows y-intercept c; $a(x - r)(x - s)$ shows zeros.

Vertex calculation

$h = -b/(2a)$; $k = f(h)$, or complete the square.

Example 1

Describe $y = -2(x - 3)^2 + 8$ and find its zeros.

Solution

1. Vertex $(3,8)$, axis $x = 3$; opens down; maximum 8.
2. Set $y = 0$: $(x - 3)^2 = 4 \rightarrow x = 1$ or 5.
3. Range $y \leq 8$; domain all real x.

Quadratic Functions and Transformations

Example 2

Convert $y = x^2 - 6x + 5$ to vertex form.

Solution

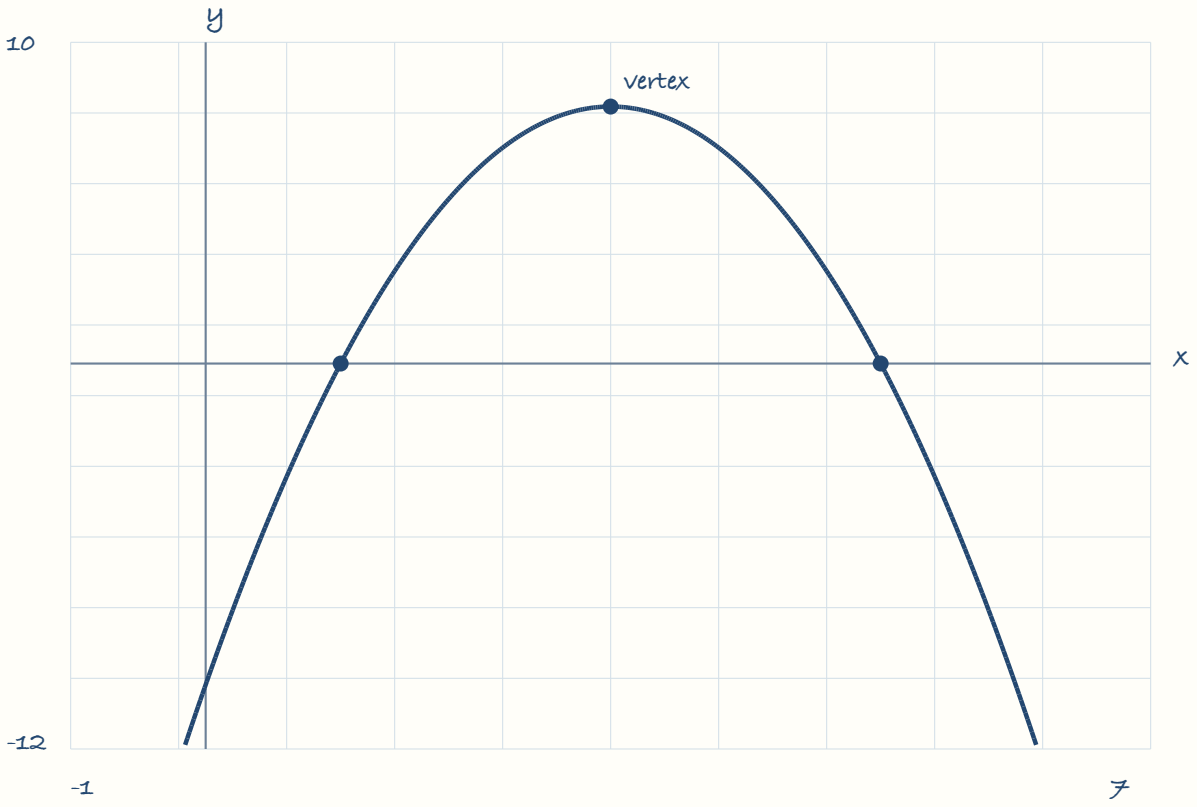
1. Complete square: $x^2 - 6x + 9 - 9 + 5$.

2. $y = (x - 3)^2 - 4$; vertex $(3,-4)$.

Quadratic Functions and Transformations

Visual notes | read the graph

$y = -2(x - 3)^2 + 8$



The zeros 1 and 5 are equally far from the axis $x = 3$. The maximum output is 8.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Quadratic Functions and Transformations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State vertex, axis and range of $y = 3(x + 1)^2 - 7$.

2. Convert $y = x^2 + 4x + 1$ to vertex form.

3. Find the y-intercept and zeros of $y = (x - 2)(x + 4)$.

4. A parabola has vertex (0,5) and passes through (2,-3). Find its equation.

Quadratic Functions and Transformations

Practice answers | check after trying

1. Vertex $(-1,-7)$, axis $x = -1$, range $y \geq -7$.

2. $(x + 2)^2 - 3$.

3. y-intercept $(0,-8)$; zeros $x = 2$ and -4.

4. $-3 = 4a + 5$ gives $a = -2$, so $y = -2x^2 + 5$.

Quadratic Equations and Applications

Concept

Quadratic equations can be solved by factoring, square roots, completing the square or a formula.

Formula

$x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$, with $a \neq 0$.

Discriminant

$D > 0$: two real roots; $D = 0$: repeated root; $D < 0$: no real roots.

Zero product

Set the equation equal to zero before setting each factor to zero.

Context

Reject impossible lengths/times; check the vertex lies in the feasible domain.

Example 1

Solve $2x^2 - 5x - 3 = 0$.

Solution

1. Factor: $(2x + 1)(x - 3) = 0$.

2. $2x + 1 = 0$ or $x - 3 = 0 \rightarrow x = -1/2$ or 3.

Quadratic Equations and Applications

Example 2

Solve $x^2 + 4x - 1 = 0$ by completing the square.

Solution

1. $x^2 + 4x = 1 \rightarrow (x + 2)^2 = 5$.
2. Take both roots: $x + 2 = \pm\sqrt{5}$.
3. $x = -2 \pm \sqrt{5}$.

Example 3

Find the maximum of $h(t) = -5t^2 + 20t + 1$ for $t \geq 0$.

Solution

1. Vertex time: $t = -20/[2(-5)] = 2$.
2. $h(2) = -20 + 40 + 1 = 21$. Maximum: 21 at $t = 2$.

Quadratic Equations and Applications

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $x^2 - 9x + 20 = 0$.

2. Solve $2(x - 1)^2 = 18$.

3. Solve $x^2 + 2x + 5 = 0$ over the reals.

4. Find the maximum of $-x^2 + 6x + 4$ for $0 \le x \le 8$.

Quadratic Equations and Applications

Practice answers | check after trying

1. $(x - 4)(x - 5) = 0$, so 4 or 5.

2. $(x - 1)^2 = 9$ gives $x = 1 \pm 3$, so -2 or 4.

3. $D = 4 - 20 = -16$, so no real roots.

4. $-(x - 3)^2 + 13$ has maximum 13 at $x = 3$, which lies in the allowed interval.

Linear and Quadratic Inequalities

Concept

An inequality describes a set of values, often one or more intervals.

Reverse sign

Reverse $<$ or $>$ only when multiplying/dividing by a negative value.

Sign chart

Find zeros, split the number line, test each interval.

Endpoints

$\leq$ or $\geq$ includes zeros; $<$ or $>$ excludes them.

Notation

(a,b) excludes endpoints; $[a,b]$ includes them; $\cup$ joins intervals.

Example 1

Solve $-3x + 7 \leq 16$.

Solution

1. $-3x \leq 9$.
2. Divide by -3 and reverse: $x \geq -3 \rightarrow [-3,\infty)$.

Linear and Quadratic Inequalities

Example 2

Solve $x^2 - 5x + 6 > 0$.

Solution

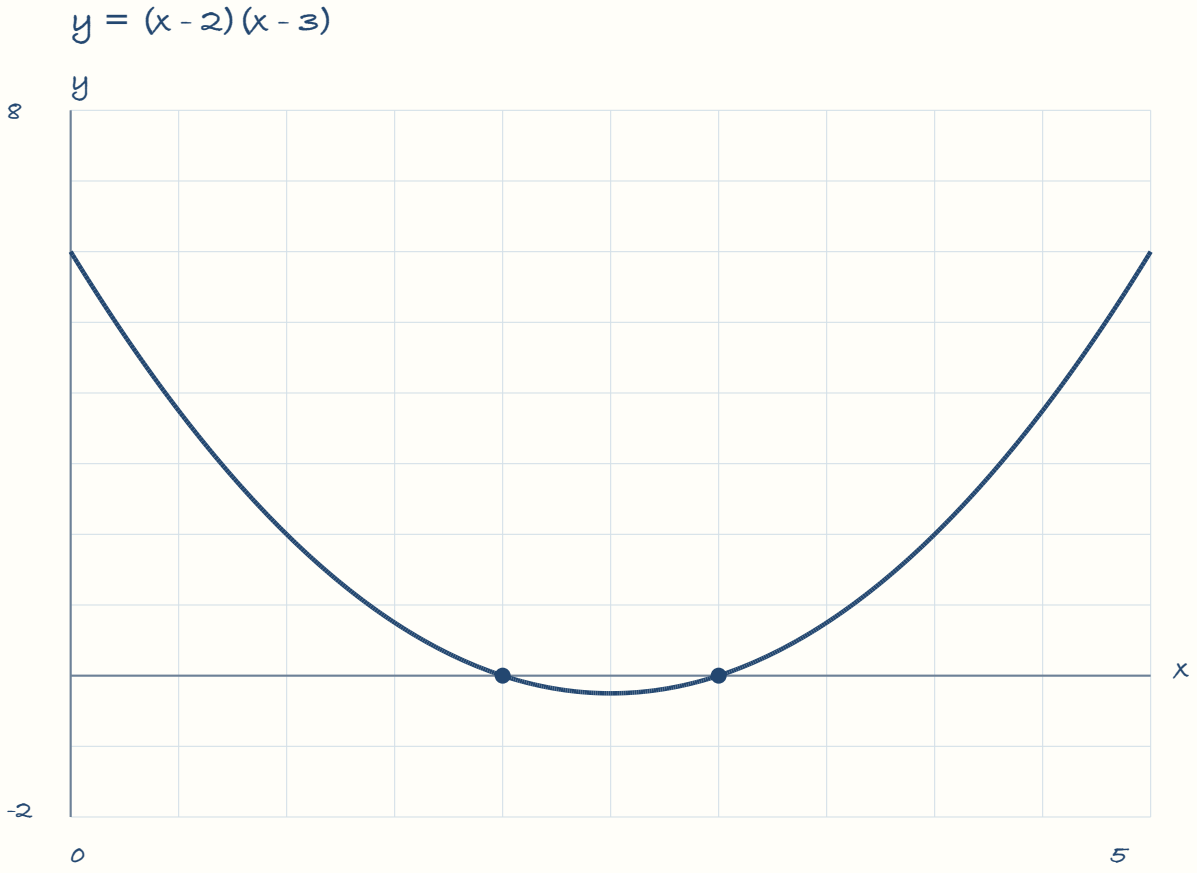
1. Factor: $(x - 2)(x - 3) > 0$. Critical values: 2,3.

2. Product signs: positive for $x < 2$; negative for $2 < x < 3$; positive for $x > 3$.

3. Answer: $(-\infty,2) \cup (3,\infty)$. Exclude the zeros.

Linear and Quadratic Inequalities

Visual notes | read the graph



Positive outside the zeros and negative between them. Strict inequalities exclude the zeros.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Linear and Quadratic Inequalities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $5 - 2x > 11$.

2. Solve $x^2 - 4 \leq 0$.

3. Solve $(x - 1)(x + 3) \geq 0$.

4. Solve $-(x - 2)^2 > 0$.

Linear and Quadratic Inequalities

Practice answers | check after trying

1. $-2x > 6$ gives $x < -3$.

2. Zeros -2 and 2; the quadratic is non-positive between them: $[-2,2]$.

3. Positive outside the roots and zero at them: $(-\infty,-3] \cup [1,\infty)$.

4. No solution. A square is non-negative, so its negative cannot be strictly positive.

Angles and Non-Right Triangles

Concept

Use the unit circle for general angles, and sine/cosine laws for non-right triangles.

Unit circle

$(\cos\theta, \sin\theta)$; $\tan\theta = \sin\theta/\cos\theta$ when defined.

Special angles

Use 30-60-90 and 45-45-90 triangles; apply quadrant signs.

Triangle laws

$a/\sin A = b/\sin B$; $c^2 = a^2 + b^2 - 2ab\cos C$.

SSA ambiguity

Check both B and $180° - B$; keep only valid triangles.

Example 1

Find $\sin150°$, $\cos150°$ and $\tan150°$.

Solution

1. Reference angle = 30°; quadrant II has positive sine, negative cosine.

2. $\sin = 1/2$; $\cos = -\sqrt{3}/2$; $\tan = -1/\sqrt{3}$.

Angles and Non-Right Triangles

Example 2

Sides 5 and 7 enclose 60°. Find the opposite side c.

Solution

1. $c^2 = 5^2 + 7^2 - 2(5)(7)\cos 60° = 39$.

2. $c = \sqrt{39} \approx 6.24$.

Example 3

Given $A = 30°$, $a = 7$ and $b = 10$, how many triangles exist?

Solution

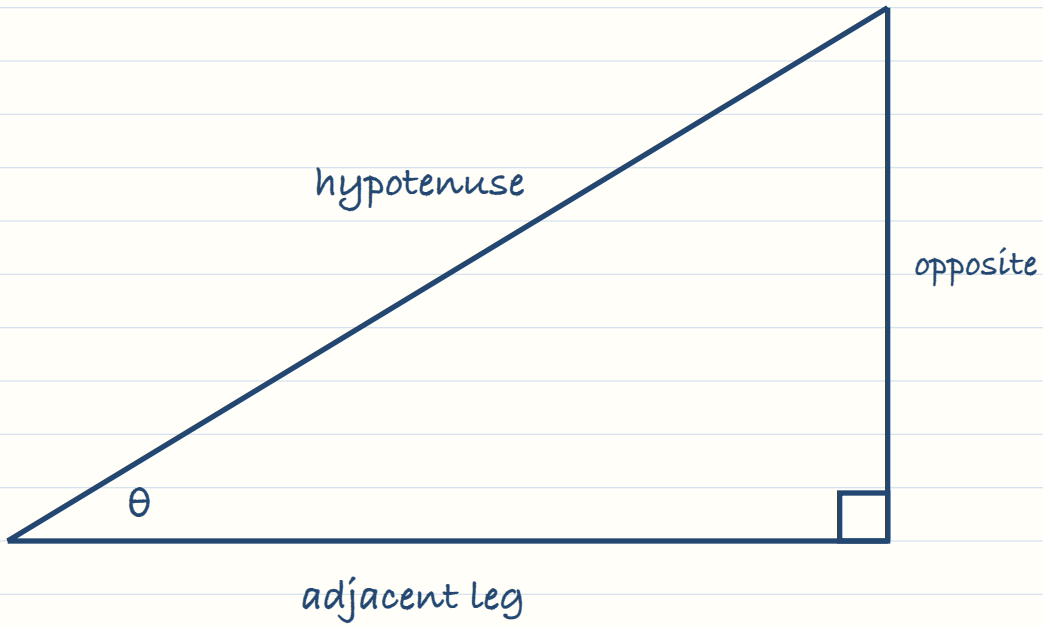
1. $\sin B = 10 \sin 30°/7 = 5/7$.

2. $B \approx 45.58°$ or $134.42°$. Both leave $A + B < 180°$.

3. Two triangles are possible.

Angles and Non-Right Triangles

Visual notes | label before calculating



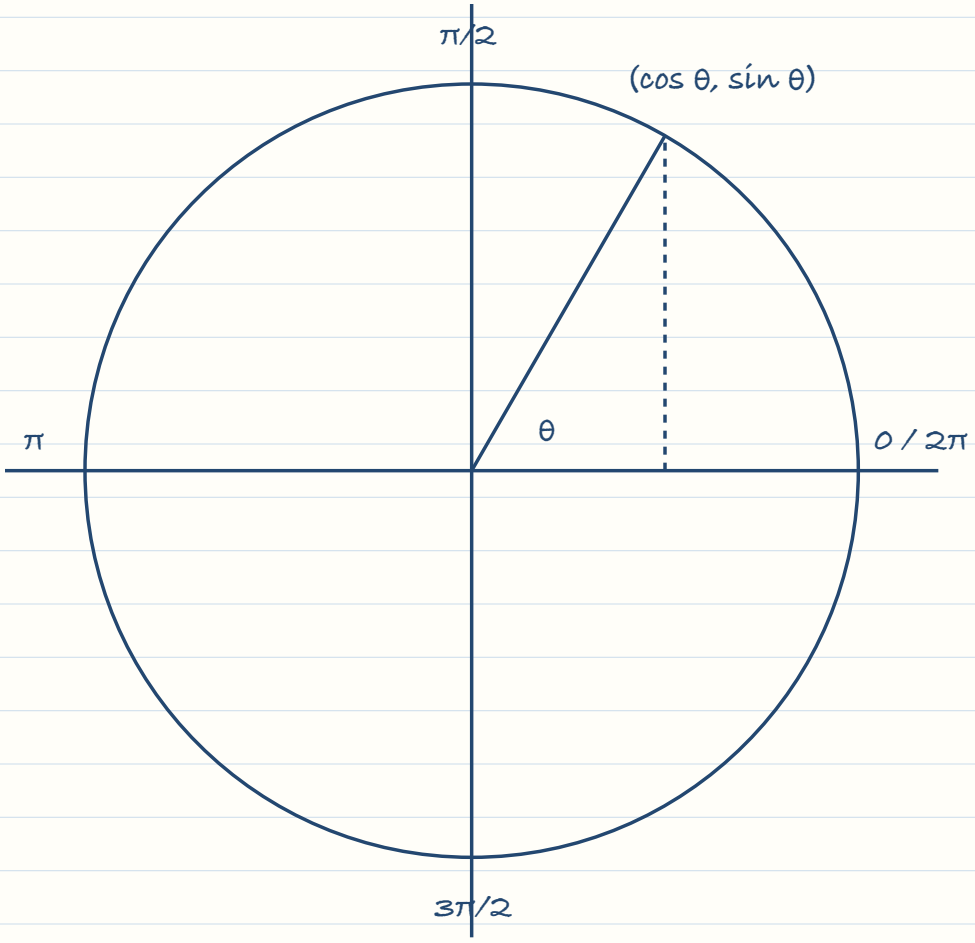
The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Pythagoras: $a^2 + b^2 = c^2$. For right triangles, sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, and tan θ = opposite/adjacent.

For a non-right triangle, instead match opposite pairs a/A, b/B and c/C. Sine law needs an opposite pair; cosine law uses an included angle or three sides. Do not apply SOH CAH TOA directly to an arbitrary non-right triangle.

Angles and Non-Right Triangles

Visual notes | unit circle



The radius is 1. The horizontal coordinate is cosine and the vertical coordinate is sine. The drawn angle is π/3 (60°), giving the exact point (1/2, √3/2).

Signs follow the coordinates: upper half means positive sine; right half means positive cosine. Tangent is sin θ/cos θ and is undefined on the vertical axis.

Angles and Non-Right Triangles

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\sin 225°$ and $\cos 225°$.

2. Solve $\cos\theta = -1/2$ on $0° \le \theta < 360°$.

3. Find c if $a = 8$, $b = 6$ and included $C = 90°$.

4. If $A = 30°$, $a = 4$ and $b = 10$, how many triangles exist?

Angles and Non-Right Triangles

Practice answers | check after trying

1. Both are $-\sqrt{2}/2$, from quadrant III with reference 45°.

2. 120° and 240°.

3. $c = \sqrt{(64 + 36)} = 10$.

4. $\sin B = 10 \sin 30°/4 = 1.25$, impossible, so no triangle.

Compound Interest, Payments and Loans

Concept

Compound interest grows a changing balance. Payment timing and rate periods must match.

Compound growth

$A = P(1 + r/m)^{mt}$; periodic rate $i = r/m$.

Deposits / loans

$B_{new} = B_{old}(1 + i) +$ deposit, or minus payment.

Regular payments

Equal end-period payments, $i>0$: $FV = D[(1+i)^{N} - 1]/i$; loan $R = Pi/[1-(1+i)^{-N}]$.

Compare costs

Check fees, timing, total payments and retained value. Rates here are hypothetical.

Example 1

Grow $2000 at 6% nominal annual interest, compounded monthly, for 2 years.

Solution

1. $i = 0.06/12 = 0.005$; $N = 24$.
2. $A = 2000(1.005)^{24} \approx \$2254.32$.
3. Interest $\approx \$254.32$.

Compound Interest, Payments and Loans

Example 2

A $1000 loan charges 1% monthly. Make a $90 end-of-month payment.

Solution

1. Interest $= 1000(0.01) = \$10$.

2. New balance $= 1000 + 10 - 90 = \$920$.

3. Principal repaid $= 90 - 10 = \$80$.

Compound Interest, Payments and Loans

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the value of $1500 after 3 years at 4% compounded annually.

2. Find the effective annual rate for 12% nominal compounded monthly.

3. A $500 balance grows 2% in a period before a $60 payment. Find the new balance.

4. Why might doubling the loan term fail to halve the total borrowing cost?

Compound Interest, Payments and Loans

Practice answers | check after trying

1. $1500(1.04)^3 = \$1687.30$ to cents.

2. $(1.01)^{12} - 1 \approx 0.126825 = 12.68\%$.

3. $500(1.02) - 60 = \$450$.

4. Interest continues to accumulate on outstanding balances for longer; lower periodic payments do not necessarily mean lower total interest.